

```

1 import components.simplereader.SimpleReader;
2
3 /**
4  * To calculate the square root.
5  *
6  * @author Eric Zhou.
7  */
8 public final class Newton3 {
9
10     /**
11      * No argument constructor--private to prevent instantiation.
12      */
13     private Newton3() {
14     }
15
16     /**
17      * Computes estimate of square root of x to within relative error 0.01%.
18      *
19      * @param x
20      *         positive number and zero to compute square root of
21      * @param epInput
22      *         the relevant error and no more than 0.01%
23      * @return estimate of square root
24      */
25     private static double sqrt(double x, double epInput) {
26         final double ep = epInput; //set the error
27         if (x == 0) { //solve the problem that x equals to zero
28             return x;
29         } else {
30             double r = x;
31             while ((Math.abs(r * r - x) / x) > ep * ep) { //calculate the square root
32                 r = (r + (x / r)) / 2;
33             }
34             return r;
35         }
36     }
37
38     /**
39      * Main method.
40      *
41      * @param args
42      *         the command line arguments
43      */
44     public static void main(String[] args) {
45         SimpleReader in = new SimpleReader1L();
46         SimpleWriter out = new SimpleWriter1L();
47         out.println("Enter a eplison: ");
48         double epInput = in.nextDouble(); //ask for the error
49         do {
50             out.println("Enter a non-negative number: "); //ask for the number
51             out.println(sqrt(in.nextDouble(), epInput));
52             out.println("Do u want to continue? Press 'y' or anything else to stop");
53         } while (in.nextLine().equals("y")); //check whether the user want to continue
54         /*
55          * Close input and output streams
56          */
57         in.close();
58         out.close();
59     }
60 }
61
62
63
64
65
66

```